Journal of Food Protection 89 (2026) 100929

Contents lists available at ScienceDirect

Journal of Food Protection

journal homepage: www.elsevier.com/locate/jfp

Research Paper

Predicting Food Safety Problems Among Catering Service Operators Using Longitudinal Regulatory Data

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ARTICLE INFO

Keywords:
Catering service inspection
Complaint and whistleblower records
Cross-period transition analysis
Food safety regulation
Machine learning
SHAP

ABSTRACT

Full-coverage on-site inspection of catering service operators is difficult to sustain when regulated entities are numerous, and risk is unevenly distributed. Using administrative regulatory data from Shanghai, China (2020–2024), we constructed a longitudinal dataset comprising 46,614 catering service operators and 150,467 inspection intervals. Machine-learning models were developed to predict whether at least one problem would be detected at the subsequent inspection, and model predictions were interpreted using SHAP. We further examined cross-period persistence and transition among categories. The selected primary model, LightGBM, achieved an AUC of 0.866, average precision of 0.806, and F1 score of 0.761 on the test set, with the highest recall among the individual models (0.805). SHAP analysis indicated that current-inspection findings and cumulative inspection history were the dominant predictors, with complaint and whistleblower records providing complementary information. Cross-period transition analysis revealed that recurrence patterns differed by problem category: environmental hygiene and raw material control problems recurred frequently but with low category specificity, whereas qualification and labeling compliance problems were rarer but showed stronger self-persistence. These findings demonstrate that routine regulatory records can support finer stratification of operators according to their probability of inspection-detected problems and more targeted follow-up by distinguishing operators with general recurrent problems from those with persistent category-specific deficiencies.

Foodborne diseases impose a substantial global burden, causing an estimated 600 million illnesses and 420,000 deaths each year (World Health Organization Regional Office for the Western Pacific, 2024; Havelaar et al., 2015). In China, national surveillance from 2003 to 2017 documented 19,517 foodborne disease outbreaks (Li et al., 2020). Catering service operators, including restaurants and institutional canteens, are consistently recognized as high-risk settings. However, their large number relative to available inspection capacity creates a structural mismatch between regulatory resources and the distribution of risk. To address this challenge, risk-based classification and tiered management have been increasingly incorporated into food-safety regulation, with inspection frequency and intensity informed by operator-specific risk factors, such as food type, production process, compliance history, and hazard severity (Ng et al., 2022; van Asselt et al., 2021). In China, risk classification has been embedded in food safety regulation since 2016, complemented by the "double-random, one-public" mechanism (State Administration for Market Regulation of China, 2021).

The effectiveness of risk-based management depends on the information available to differentiate operator risk. Administrative inspection records remain central to this process because they are systematically documented and directly linked to regulatory decisions. However, they are generated only when inspections occur and may not fully capture routine operating conditions. In particular, preannounced inspections may allow food businesses to correct noncompliances before the inspection, potentially obscuring routine operating conditions (Kaskela et al., 2021). Consumer-generated data, including online reviews and social media, have been explored for detecting unreported foodborne illness and supporting foodborne disease surveillance (Harrison et al., 2014; Harris et al., 2014; Schomberg et al., 2016). Formal complaint systems have also been used to identify food-safety events associated with catering service operators (Kim et al., 2024). Formally registered complaint records provide structured information on complaint category, frequency, and timing, but their use in predicting inspection-detected noncompliance remains underexplored.

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https://doi.org/10.1016/j.jfp.2026.100929
Received 1 July 2026; Accepted 20 September 2026
Available online 22 September 2026
0362-028X/© 2026 The Author(s). Published by Elsevier Inc. on behalf of International Association for Food Protection.
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Machine-learning and data-science methods have increasingly been integrated into food-safety monitoring, prediction, and decision-support systems across the food chain (Nychas et al., 2021), with recent reviews further documenting their expanding role in food safety risk assessment (Zhang et al., 2025). In parallel, computational platforms such as Microrisk Lab have facilitated parameter estimation and deterministic or stochastic simulation of microbial behavior, illustrating the broader contribution of quantitative predictive modeling to food-safety decision support (Liu et al., 2021). Predictive approaches have also been applied specifically to inspection prioritization, including text-based violation prediction (Kang et al., 2013), analyses of the Chicago food-inspection forecasting model (Kannan et al., 2019), and subsequent work addressing more equitable allocation of inspection resources (Singh et al., 2022). Other studies have predicted noncompliant outlets (Oldroyd et al., 2021) and developed market-level or structured visual risk-scoring analysis (Gao et al., 2022; Chen et al., 2023). Recent studies have increasingly shifted their focus toward temporal validity and model explainability in inspection prediction (Cai & Tsai, 2026; Sari et al., 2026). Cai and Tsai (2026) developed a temporally valid, leakage-free framework integrating current inspection data, longitudinal history, and historical narratives for prospective high-risk screening. Sari et al. (2026) demonstrated the use of an explainable transformer-based model for pathogen-oriented food-safety inspection-grade prediction. Together, these studies underscore the importance of temporally valid and interpretable approaches for translating routine inspection data into actionable risk signals.

Predicting whether a problem will occur addresses where to inspect, but regulatory follow-up also requires understanding how identified problem categories behave over time. Previous studies have documented recurrent critical violations in catering service operators (Phillips et al., 2006) and associations between prior inspection findings and subsequent compliance (Kaskela et al., 2021). Repeated interactions between inspectors and catering service operators may also influence detected violations, further supporting the importance of a longitudinal regulatory perspective (Jin & Lee, 2018). However, limited research has systematically examined the persistence and transition of inspection-detected problem categories across consecutive inspection periods. Transition probability matrices provide a natural framework for characterizing such cross-period dynamics, while the lift statistic—the ratio of observed transition probability to the expected probability under independence—can identify transitions that occur more frequently than expected by chance (Hahsler & Hornik, 2007). Such temporal patterns may reveal differences in the dynamics of specific problem categories.

To address these gaps, this study used administrative regulatory data from the Shanghai Municipal Administration for Market Regulation covering the period 2020–2024, comprising 46,614 catering service operators and 150,467 inspection intervals. The analysis proceeded along two complementary dimensions. First, we predicted future inspection-detected problems using current-inspection signals, inspection-history features, complaint and whistleblower records, and operator attributes, with SHAP used to identify key predictors. Second, we examined cross-period persistence and transitions among inspection problem categories using transition probability matrices and lift statistics. The overall study design and analytical workflow are presented in Figure 1.

Materials and methods

Data source, study design, outcome, and predictors. This study used administrative food safety regulatory data from the Shanghai Municipal Administration for Market Regulation covering January 1, 2020 to December 31, 2024. Because the same catering service operator could be inspected multiple times during the study period, the data had a longitudinal repeated-event structure. We defined the analytical unit as the enterprise-interval, representing the period between two consecutive inspections of the same operator. This unit preserves temporal ordering and supports dynamic prediction: for each interval, the outcome was defined as whether at least one problem was recorded at the subsequent inspection, while predictors were constructed using only information available before that subsequent inspection. Specifically, current-inspection signals referred to findings recorded at the inspection that opened the interval, and complaint and whistleblower features were updated using accepted records accumulated up to the subsequent inspection. Candidate predictors were grouped into five domains (Table 1): regulatory inspection history (e.g., cumulative number of inspections and historical problem rate), current-inspection signals (e.g., whether problems were detected at the interval-opening inspection), complaint and whistleblower history (e.g., cumulative accepted complaint counts by category), recent complaint dynamics (e.g., complaint frequency and category composition in the most recent window), and enterprise baseline attributes (e.g., operator scale, license category, and district). The outcome was whether at least one problem was recorded at the subsequent inspection, selected because it reflects a standardized regulatory finding and corresponds to the practical decision of identifying operators that may require closer follow-up. The outcome should be interpreted as an administratively observed regulatory endpoint rather than a direct measure of latent food-safety hazard, foodborne illness risk, or legal violation severity. Operators that ceased operation before December 31, 2024, and therefore did not receive a subsequent inspection were treated as having terminated follow-up at their last observed inspection. Records with missing inspection status or enterprise identifiers were excluded, as were observations with abnormal interval lengths or without an observed subsequent inspection outcome. After screening, the final analytical dataset comprised 46,614 catering service operators and 150,467 inspection intervals.

Data preprocessing. Data preprocessing included skewness correction, standardization, and categorical encoding. Highly right-skewed variables—including cumulative inspection counts, problem counts, complaint and whistleblower counts, and interval duration—were log-transformed using $\log1p(x)$. Continuous predictors were then Z-score standardized. Nominal categorical variables, including administrative district, inspection type, and business mode, were retained as categorical inputs using native categorical feature handling. These preprocessing steps were applied uniformly across all analyses to maintain consistency; for tree-based models, monotone transformations and standardization do not alter split decisions and therefore do not affect predictive performance. All analyses were conducted in Python 3.10.

Descriptive and univariate analyses. Descriptive analyses were stratified by the outcome (problem detected vs. no problem detected at the subsequent inspection). Quantitative variables were summarized as mean ± standard deviation, and categorical variables were summarized as counts and percentages. Between-group differences were evaluated using the Mann–Whitney U test for quantitative variables and the chi-square test or Fisher's exact test for categorical variables. Effect sizes were reported as standardized mean differences for quantitative and binary variables and Cramér's V for multicategory variables.

Model development and validation. All 38 candidate predictors listed in Table 1 were retained for model development. These prespecified variables were derived from routinely collected complaint, whistleblower, inspection, and enterprise-registration records and represented interpretable regulatory attributes based on domain knowledge and regulatory practice. We chose not to perform data-driven variable selection because it may introduce additional uncertainty into the final model (Heinze et al., 2018). To assess whether the training sample was sufficient to support reliable model development, we applied the criteria proposed by Riley et al. (2019, 2020). The results supported the adequacy of the training sample (Table S1).

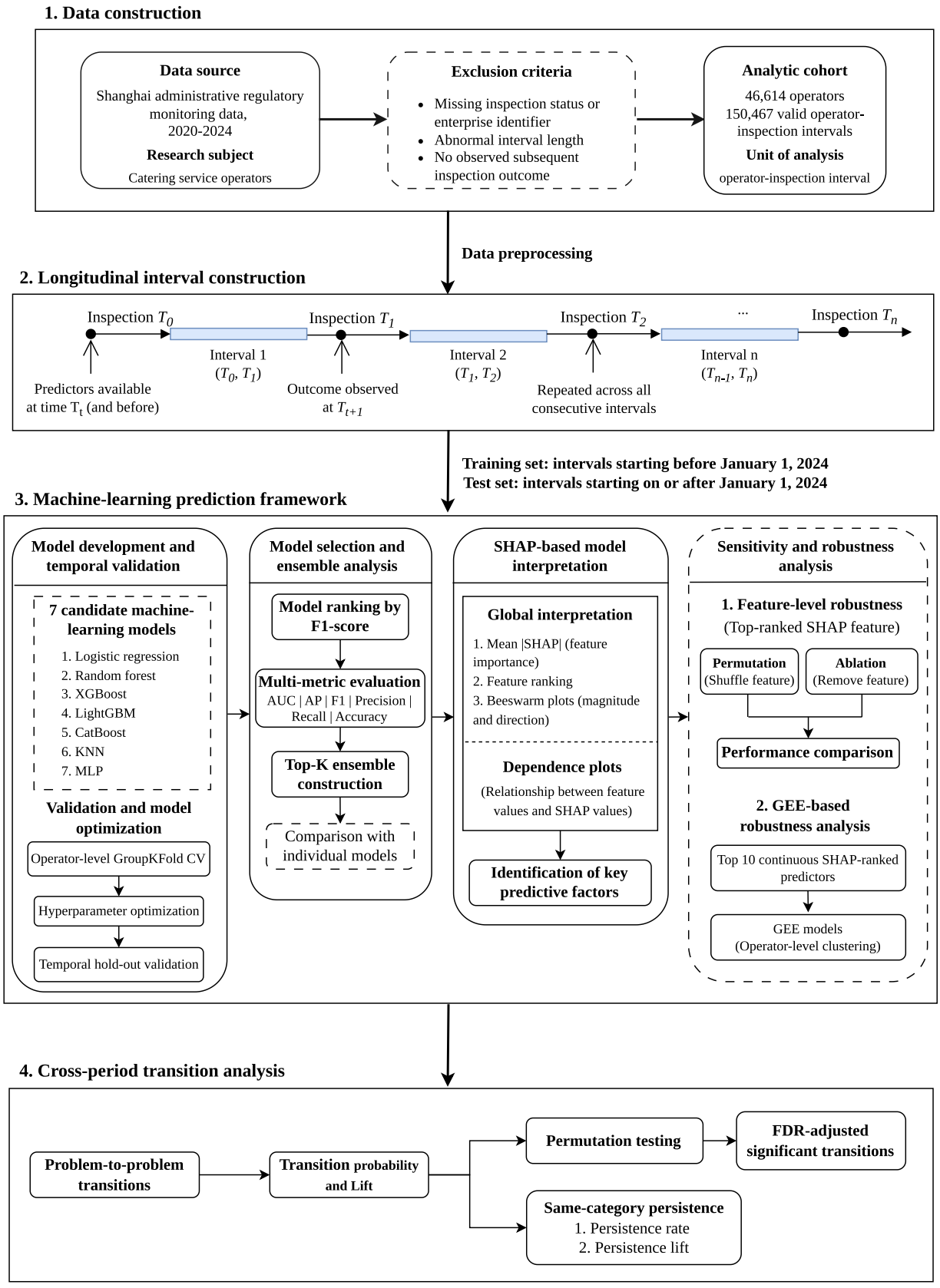


Figure 1. Overall study design and analytical workflow. Shanghai administrative regulatory data (2020–2024) were processed to construct longitudinal intervals between consecutive inspections of the same operator. The workflow comprises four components: data construction, longitudinal interval construction, machine-learning prediction, and cross-period transition analysis. The prediction framework includes model development and temporal validation, model selection and ensemble analysis, SHAP interpretation, and sensitivity and robustness analyses using permutation, ablation, and GEE. Cross-period analysis characterizes problem-category transitions and same-category persistence using transition probabilities, persistence rates, and Lift, with permutation testing and false discovery rate adjustment for transition significance.

Table 1
Feature domains, subdomains, and definitions of candidate predictors

Feature domain	Subdomain	Feature	Measurement type	definition
Regulatory inspection	Historical cumulative	Days since last inspection	Continuous duration	/
		Historical inspection count	Count	/
		Historical violation rate	Proportion	Number of prior inspections with at least one recorded problem divided by the total number of prior inspections.
		Historical violation count	Count	/
		Consecutive violation count	Discrete count	Number of most recent consecutive prior inspections with at least one recorded problem, counting backward from the current inspection.
		Historical problem count	Count	/
		Historical rectification completion rate	Proportion	Number of historical problem items with completed rectification divided by the total number of historical inspections with problem items.
		Historical on-site rectification rate	Proportion	Number of historical problem items rectified on site divided by the total number of historical inspections with problem items.
	Current inspection	Current problem-type diversity	Discrete count	/
		Inspection interval days	Continuous duration	Days between the current and subsequent inspections. Used in descriptive and GEE analyses only; excluded from the machine-learning feature set.
		Current main problem category	Nominal categorical	Dominant problem category at the current inspection based on item count, and inspections without problems retained as an explicit no-record category.
		Current problem count	Count	/
		Current main problem category ratio	Proportion	Number of problem items in the dominant category divided by the total number of problem items at the current inspection; set to 0 when no problem was recorded.
		Inspection type	Nominal categorical	/
		Has inspection summary	Binary	/
		Has rectification order	Binary	/
Complaint and whistleblower	Historical cumulative	Historical complaint count	Count	/
		Complaint type diversity	Count	/
		Complaint ratio in complaint/ whistleblower records	Proportion	Number of consumer complaint records divided by the total number of accepted complaint and whistleblower records.
		Historical main complaint object category	Nominal categorical	Most frequent complained-about object category in the cumulative record; no-record retained as explicit category.
		Whistleblower report count	Count	/
		Historical complaint issue category diversity	Discrete count	/
		Historical main complaint issue category	Nominal categorical	Most frequent complaint issue category in the cumulative record; no-record retained as explicit category.
		Historical main complaint issue category count	Count	/
		Historical complaint object category diversity	Discrete count	/
		Historical main complaint object category count	Count	/
	Recent dynamic	Days since last complaint	Continuous duration	Days from the most recent accepted complaint/ whistleblower record to the interval start; enterprises without prior records were assigned a sentinel value of 9999.
		Complaints in past year	Count	/
		Past-year main complaint issue category	Nominal categorical	/
		Past-year main complaint issue category count	Count	/
Enterprise baseline attributes	Administrative context	Administrative district	Nominal categorical	/
	Business scope and category	Primary business scope	Nominal categorical	/
		Secondary business scope	Nominal categorical	/
		Business scope count	Discrete count	/
		Business scope category count	Discrete count	Number of distinct licensed business-scope categories derived from operating-license codes.
		Business mode	Nominal categorical	/
	Operational scale and age	Years in operation	Continuous duration	Years from enterprise registration date to the interval start.
		Registered capital	Continuous	/
		Employee count	Count	/

Note. Features are organized into three domains: regulatory inspection, complaint and whistleblower, and enterprise baseline attributes. Each domain is further divided into subdomains reflecting the temporal scope of the information (historical cumulative, current inspection, recent dynamic) or the attribute type (administrative context, business scope and category, operational scale and age). Measurement type classifies variables by their substantive scale: count, proportion, continuous duration, continuous monetary value, binary, discrete count, or nominal categorical. "/" in the operational definition column indicates that the variable meaning is self-evident from the feature name.

A time-based split was used for model development and validation. Intervals starting before January 1, 2024 were assigned to the training set, and intervals starting on or after January 1, 2024 were assigned to the test set. This forward-looking partition preserved temporal order and reduced look-ahead bias (Roberts et al., 2017). Within the training set, fivefold grouped cross-validation was performed using GroupKFold with the enterprise identifier as the grouping key, ensuring that all intervals from the same operator appeared in only one fold. Seven classification algorithms were evaluated: logistic regression, random forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), CatBoost (Prokhorenkova et al., 2018), k-nearest neighbors, and multilayer perceptron. Hyperparameters were optimized using the tree-structured Parzen estimator (TPE), a sequential model-based optimization method that uses the results of previous evaluations to guide subsequent hyperparameter selection (Bergstra et al., 2011). TPE was implemented in Optuna (Akiba et al., 2019), with the mean F1 score from fivefold operator-grouped cross-validation within the training set as the optimization objective. F1 score was chosen because it balances precision and recall in a regulatory context where both missed detections and excessive false alarms carry practical costs. The search spaces covered the main complexity, regularization, sampling, and imbalance-handling parameters of each algorithm, as detailed in Table S2.

After hyperparameter selection, base learners were fitted and calibrated using isotonic regression through fivefold operator-grouped cross-validation, with calibrated probabilities averaged across folds (Zadrozny & Elkan, 2002). Decision thresholds for individual models and ensembles were selected by maximizing test-set F1 over thresholds from 0.05 to 0.94 in increments of 0.01. LightGBM was retained as the primary model based on its comparative performance and high recall. Posthoc Top-K weighted ensembles (K = 2–5) used members ranked by test-set F1 and normalized weights proportional to their stored training-set cross-validation F1 scores, aligning weighting with the optimization objective (Dietterich, 2000; Table S2). Final model performance was evaluated on the test set using AUC, average precision, F1 score, precision, recall, accuracy, and confusion-matrix results.

SHAP-based model interpretation and robustness analysis. SHapley Additive Explanations (SHAP) provides a framework for interpreting model predictions (Lundberg & Lee, 2017; Lundberg et al., 2020), with applications across different machine-learning models, including Random Forest (Leung et al., 2022; Nychas et al., 2025). We used SHAP to examine how individual predictors contributed to predicted inspection-problem risk. Global SHAP importance was quantified as the mean absolute SHAP value for each feature. SHAP beeswarm plots were used to visualize the overall magnitude and direction of feature effects across the test set, and SHAP dependence plots were used to examine how selected feature values related to their model contributions.

To assess whether the selected model relied disproportionately on its most influential predictor, we conducted two complementary sensitivity analyses targeting the feature with the highest mean absolute SHAP value. In the permutation analysis, the feature values were randomly permuted in the test set while the fitted model and all other predictors were held unchanged. In the ablation analysis, the feature was removed and LightGBM was retrained on the same training data using the original hyperparameters, then evaluated on the test set. Both analyses were evaluated using the same metrics as the original model.

Because the analytical dataset contained repeated inspection intervals from the same enterprise, robustness analyses assessed whether the main SHAP-identified predictors remained associated with the inspection problem outcome after accounting for within-enterprise correlation. Generalized estimating equation models were fitted with the enterprise identifier as the clustering unit, a logit link function, binomial variance, and robust sandwich standard errors (Liang & Zeger, 1986; Ballinger, 2004). The primary sensitivity model used an exchangeable working correlation structure, and an independence working-correlation model was fitted as a comparator. The models included the top ten continuous or ordinal predictors ranked by global SHAP importance, excluding nominal categorical variables whose multilevel encodings do not represent ordinal or continuous variation and therefore cannot be meaningfully summarized as single odds ratios. Robustness was judged by the direction, magnitude, and statistical significance of associations across working-correlation structures rather than by treating the GEE models as competing prediction models.

Cross-period transition analysis. To characterize the temporal structure of regulatory findings, we examined cross-period persistence and transitions among inspection problem categories across consecutive inspections of the same operator. For each pair of consecutive inspections where both recorded at least one problem, antecedent–consequent category pairs were constructed from problem categories observed at the earlier and subsequent inspections, respectively. These pairs were used to generate transition count matrices and row-normalized conditional probability matrices. Observation-level persistence analysis also included observations with no recorded problem category at the subsequent inspection.

To account for differences in category frequency, Lift was calculated at the expanded category-pair level by dividing each row-normalized transition proportion by the marginal proportion of the subsequent category among all expanded category pairs (Hahsler & Hornik, 2007). Values greater than 1 indicate enrichment relative to independence. Statistical significance was assessed using 1,000 one-sided permutation tests in which the subsequent-inspection category sets were randomly shuffled across observations while preserving the within-inspection category structure. *P* values were adjusted using the Benjamini–Hochberg false discovery rate procedure (Benjamini & Hochberg, 1995). For each category, the persistence rate was the proportion of observations containing that category at the current inspection that also contained it at the subsequent inspection. Persistence Lift was calculated by dividing this rate by the proportion of all analytical observations containing that category at the subsequent inspection.

Results

Baseline characteristics. Baseline comparisons showed that intervals with at least one problem detected at the subsequent inspection differed from intervals with no detected problems mainly in current and historical inspection-related features (Table 2). Because of the large sample size, interpretation focused on effect-size magnitude rather than statistical significance alone. Variables were retained in Table 2. If they showed a standardized mean difference exceeding 0.10 for quantitative and binary variables or a Cramér's V exceeding 0.05 for multicategory variables, inspection type, business mode, registered capital, years in operation, employee count, and scope category count were additionally retained as key descriptors of the study population regardless of effect size.

The largest group differences were observed for current-inspection features. Current main problem category ratio (SMD = 1.199), current problem-type diversity (SMD = 1.070), and current problem count (SMD = 0.945) all indicated substantially higher problem burden at the interval-opening inspection among intervals that subsequently had problems detected. Process indicators followed the same pattern: rectification orders were more common (9.1% vs. 2.0%; SMD = 0.317) and inspection summaries more frequently recorded (99.8% vs. 96.9%; SMD = 0.227) in the problem-detected group.

Historical inspection variables showed a consistent but attenuated pattern, with historical rectification completion rate (SMD = 0.691), on-site rectification rate (SMD = 0.595), and historical problem count (SMD = 0.497) showing moderate separation. Historical violation rate (SMD = 0.269), consecutive violation count (SMD = 0.227), and historical violation count (SMD = 0.225) also differed between groups.

Table 2
Baseline characteristics of inspection intervals stratified by subsequent inspection outcome

Variable	Category / summary	Overall (n = 150,467)	No problem detected (n = 110,438)	Problem detected (n = 40,029)	Statistic	P value	Effect size
Quantitative variables, mean ± SD							
Days since last inspection	Mean ± SD	198.01 ± 195.90	189.91 ± 195.32	220.34 ± 195.75	1,941,878,684.0	<0.001	SMD = 0.156
Historical violation rate	Mean ± SD	0.03 ± 0.13	0.02 ± 0.11	0.06 ± 0.18	1,983,439,279.0	<0.001	SMD = 0.269
Historical violation count	Mean ± SD	0.15 ± 0.83	0.09 ± 0.55	0.31 ± 1.32	1,982,689,742.0	<0.001	SMD = 0.225
Consecutive violation count	Mean ± SD	0.05 ± 0.32	0.03 ± 0.21	0.12 ± 0.51	2,086,742,554.0	<0.001	SMD = 0.227
Historical problem count	Mean ± SD	0.53 ± 1.80	0.25 ± 1.14	1.31 ± 2.78	1,375,452,860.5	<0.001	SMD = 0.497
Historical rectification completion rate	Mean ± SD	0.25 ± 0.58	0.14 ± 0.46	0.57 ± 0.75	1,451,681,802.5	<0.001	SMD = 0.691
Historical on-site rectification rate	Mean ± SD	0.12 ± 0.37	0.06 ± 0.26	0.30 ± 0.52	1,673,177,351.0	<0.001	SMD = 0.595
Inspection interval days	Mean ± SD	228.99 ± 232.62	222.64 ± 235.42	246.51 ± 223.78	1,983,141,810.0	<0.001	SMD = 0.104
Current problem count	Mean ± SD	0.28 ± 0.64	0.11 ± 0.43	0.75 ± 0.85	1,114,762,259.5	<0.001	SMD = 0.945
Current problem-type diversity	Mean ± SD	0.26 ± 0.53	0.10 ± 0.36	0.68 ± 0.68	1,114,531,462.0	<0.001	SMD = 1.070
Current main problem category ratio	Mean ± SD	0.20 ± 0.39	0.08 ± 0.26	0.54 ± 0.48	1,111,435,110.0	<0.001	SMD = 1.199
Historical complaint count	Mean ± SD	0.98 ± 22.86	1.12 ± 25.93	0.60 ± 10.49	2,224,588,414.5	0.007	SMD = 0.026
Scope category count	Mean ± SD	1.55 ± 0.91	1.56 ± 0.94	1.52 ± 0.82	2,210,616,057.0	0.014	SMD = 0.043
Years in operation	Mean ± SD	10.95 ± 10.68	11.02 ± 11.98	10.74 ± 5.75	2,277,494,242.0	<0.001	SMD = 0.029
Registered capital	Mean ± SD	4.68 ± 1.46	4.70 ± 1.46	4.65 ± 1.48	2,243,819,252.5	<0.001	SMD = 0.029
Employee count	Mean ± SD	27.00 ± 461.13	25.87 ± 462.17	30.12 ± 458.23	2,185,714,120.0	<0.001	SMD = 0.009
Qualitative variables, n (%)							
Has inspection summary	Yes	146,911 (97.6)	106,975 (96.9)	39,936 (99.8)	1,072.0	<0.001	SMD = 0.227
	No	3,556 (2.4)	3,463 (3.1)	93 (0.2)			
Has rectification order	Yes	5,820 (3.9)	2,170 (2.0)	3,650 (9.1)	4,041.4	<0.001	SMD = 0.317
	No	144,647 (96.1)	108,268 (98.0)	36,379 (90.9)			
Has historical complaint	Yes	31,568 (21.0)	23,293 (21.1)	8,275 (20.7)	3.085	0.079	SMD = 0.010
	No	118,899 (79.0)	87,145 (78.9)	31,754 (79.3)			
Current main problem category	Raw material control	12,666 (8.4)	3,583 (3.2)	9,083 (22.7)	42,498.3	<0.001	Cramer's V = 0.531
	Environmental hygiene	11,022 (7.3)	2,874 (2.6)	8,148 (20.4)			
	Management systems	8,321 (5.5)	2,721 (2.5)	5,600 (14.0)			
	Quality control	473 (0.3)	182 (0.2)	291 (0.7)			
	Qualification compliance	176 (0.1)	77 (0.1)	99 (0.2)			
	Labeling compliance	108 (0.1)	62 (0.1)	46 (0.1)			
	No recorded	117,701 (78.2)	100,939 (91.4)	16,762 (41.9)			
Inspection type	Routine inspection	146,093 (97.1)	106,673 (96.6)	39,420 (98.5)	382.373	<0.001	Cramer's V = 0.050
	Special inspection	3,333 (2.2)	2,911 (2.6)	422 (1.1)			
	Unannounced inspection	1,041 (0.7)	854 (0.8)	187 (0.5)			
Business mode	Restaurant/catering service	93,722 (62.3)	67,432 (61.1)	26,290 (65.7)	546.666	<0.001	Cramer's V = 0.060
	Institutional canteen	13,120 (8.7)	9,201 (8.3)	3,919 (9.8)			
	Other	43,625 (29.0)	33,805 (30.6)	9,820 (24.5)			

Note: Data are mean ± SD for quantitative variables and n (%) for qualitative variables. Between-group differences were evaluated using the Mann–Whitney U test for quantitative variables and the chi-square test for categorical variables. Effect sizes are reported as absolute standardized mean differences (SMD) for quantitative and binary variables and Cramér's V for multicategory variables. Variables were retained if SMD > 0.10 or Cramér's V > 0.05; inspection type, business mode, registered capital, years in operation, employee count, and scope category count were additionally retained as population-level descriptors regardless of effect size.

Timing variables contributed smaller effect sizes (days since last inspection: SMD = 0.156; interval length: SMD = 0.104).

Complaint and whistleblower variables showed weak and inconsistent separation. Most complaint-related effect sizes were below 0.06, and the proportion with any historical complaint was nearly identical between groups (20.7% vs. 21.1%; SMD = 0.010). Notably, several complaint variables showed slightly higher values in the no-problem group, possibly reflecting that operators with more complaint exposure had already received regulatory attention or corrective follow-up.

Among categorical variables, current main problem category showed the strongest separation (Cramér's V = 0.531), with 91.4% of no-problem intervals having no recorded current problem compared with 41.9% of problem-detected intervals. Inspection type and business mode were statistically significant but with small effect sizes (Cramér's V = 0.050 and 0.060, respectively), indicating that the study population included a mix of inspection modes and operator types without substantial confounding by these design features. Enterprise baseline attributes—registered capital, years in operation, employee count, and scope category count—showed minimal group differences (SMD = 0.029, 0.029, 0.009, and 0.043, respectively), suggesting that future inspection outcomes were largely independent of operator scale and business scope.

Overall, baseline differences indicated that future inspection outcomes were most strongly associated with contemporaneous and historical inspection findings, while complaint-related variables contributed weaker but potentially complementary information, and enterprise baseline attributes showed negligible univariate separation.

Prediction of future inspection problems. All seven candidate algorithms showed good discrimination on the test set (Table 3). Among single models, AUC values ranged from 0.857 to 0.867, and F1 scores ranged from 0.748 to 0.761. LightGBM achieved an AUC of 0.866, average precision of 0.806, and F1 score of 0.761, with the highest recall among single models (0.805). CatBoost showed a similar F1 score (0.761) and slightly higher precision than LightGBM (0.741 vs. 0.722), but its recall was lower (0.781 vs. 0.805). Given that failing to identify a problem operator may have substantially greater consequences than conducting an unnecessary follow-up inspection in our regulatory setting, we selected LightGBM as the primary model. At

Table 3
Predictive performance of candidate models on the test set

Model	AUC	AP	F1	Precision	Recall	Accuracy	Best Threshold
LightGBM	***0.866***	***0.806***	***0.761***	***0.722***	***0.805***	***0.804***	***0.430***
TOP3	0.868	0.811	0.765	0.732	0.800	0.809	0.430
TOP2	0.867	0.808	0.764	0.734	0.798	0.809	0.440
TOP5	0.870	0.812	0.764	0.723	0.810	0.806	0.410
TOP4	0.869	0.812	0.763	0.733	0.796	0.809	0.430
CatBoost	0.865	0.804	0.761	0.741	0.781	0.809	0.460
RF	0.867	0.808	0.760	0.729	0.794	0.806	0.420
MLP	0.866	0.806	0.759	0.736	0.783	0.807	0.440
LR	0.857	0.768	0.756	0.741	0.773	0.807	0.490
XGBoost	0.862	0.798	0.755	0.715	0.801	0.799	0.420
KNN	0.858	0.787	0.748	0.710	0.791	0.794	0.380

Note. AUC, area under the receiver operating characteristic curve; AP, average precision. F1, precision, recall, and accuracy were computed at the model-specific optimal threshold. TOP-k denotes a weighted ensemble of the top-k base learners ranked by test-set F1, with weights obtained by normalizing their cross-validated F1 scores within the training set. RF, random forest; MLP, multilayer perceptron; LR, logistic regression. Models are presented in descending order of test-set F1 score; the best single model is highlighted in bold.

the selected probability threshold of 0.430, the model correctly identified 80.5% of intervals with subsequent problems (recall), while maintaining a false positive rate of 19.6% (Fig. 2c). The selected decision threshold therefore prioritized sensitivity to subsequent problems while accepting a moderate false-positive rate, consistent with the regulatory objective of minimizing missed problem operators. The histogram-based, leaf-wise training of LightGBM is also computationally efficient (Ke et al., 2017), which is advantageous for periodic model updating as new inspection records accumulate. Post hoc weighted ensembles provided only modest additional improvement (F1 = 0.763–0.765) and were therefore treated as supplementary benchmarks. The specific ensemble weighting parameters are provided in Table S2.

SHAP interpretation of inspection-problem predictions. SHAP analysis identified a concentrated set of regulatory signals driving inspection-problem predictions (Fig. 3a). The highest global contribution was observed for current problem-type diversity (mean |SHAP| = 0.677), followed by administrative district (0.385), historical problem count (0.288), current main problem category (0.272), days since the last inspection (0.198), historical inspection count (0.177), primary business category (0.127), historical rectification completion rate (0.112), and historical on-site rectification rate (0.105). The model relied most heavily on the breadth of current inspection findings, accumulated historical problem burden, inspection timing, and rectification history.

Administrative district ranked second in global importance, likely reflecting systematic differences in regulatory intensity, catering density, or local compliance environments across districts rather than a single interpretable risk factor. For this and other nominal categorical variables, SHAP beeswarm color gradients (Fig. 3b) reflect encoded numeric values rather than ordinal relationships; category-specific contributions are shown in the dependence plots (Fig. 4).

SHAP dependence plots illustrated feature-specific contribution patterns among the leading predictors (Fig. 4). Current problem-type diversity showed higher SHAP contributions for observations with at least one recorded problem category than for those with none (Fig. 4a). Historical problem count showed a monotone increasing pattern with diminishing marginal contribution at higher values (Fig. 4c). Days since the last inspection exhibited a nonlinear pattern—moderate intervals contributed positively while very short or very long intervals showed attenuated or variable effects (Fig. 4e). Current main problem category showed marked heterogeneity: raw material control, environmental hygiene, and management-system categories contributed more strongly to predicted risk than the no-record category (Fig. 4d).

Complaint and whistleblower variables were notably absent from the top global SHAP features, indicating that once inspection-history signals were included, structured complaint records provided complementary but not dominant predictive information. This is substantively consistent with the nature of these data sources: formal inspection findings directly measure regulatory noncompliance, whereas complaint records reflect consumer-facing exposure and reporting behavior.

To assess whether the prominence of Current Problem-Type Diversity in the SHAP analysis reflected over-reliance on a single predictor, we performed two complementary sensitivity analyses (Table S3). Permuting this feature while holding the fitted model unchanged reduced

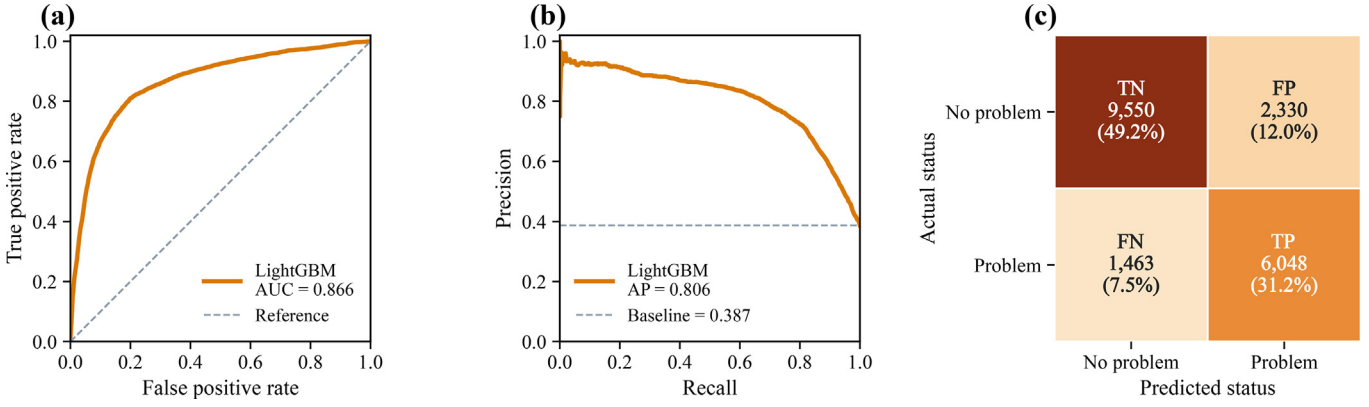


Figure 2. Performance of the LightGBM model on the test set. (a) Receiver operating characteristic curve; AUC = 0.866. (b) Precision–recall curve; AP = 0.806; the dashed line indicates the baseline prevalence (0.387). (c) Confusion matrix at the optimal threshold (0.430).

AUC from 0.866 to 0.794 and F1 from 0.761 to 0.678, indicating that the fitted model made substantial use of the information carried by this feature. In contrast, removing the feature and refitting LightGBM using the same hyperparameters and data split left performance essentially unchanged, with AUC changing only from 0.866 to 0.865 and F1 from 0.761 to 0.760. These findings indicate that Current Problem-Type Diversity was important to the fitted model, but that overlapping predictive information could be recovered from the remaining predictors after model retraining.

Temporal persistence and transitions among inspection problem categories. We examined problem-category transitions and same-category persistence across consecutive inspections (Fig. 5). Figure 5a–b summarizes expanded category pairs from inspections with recorded categories at both time points, whereas Figure 5c–d summarizes observation-level persistence, including observations with no recorded category at the subsequent inspection.

The transition probability matrix (Fig. 5a) showed that all problem categories exhibited some degree of self-transition. Environmental hygiene had the highest self-transition probability (58.7%), followed by raw material control (45.3%) and management systems (39.3%). The most frequent cross-category transitions were from raw material control to environmental hygiene and from management systems to

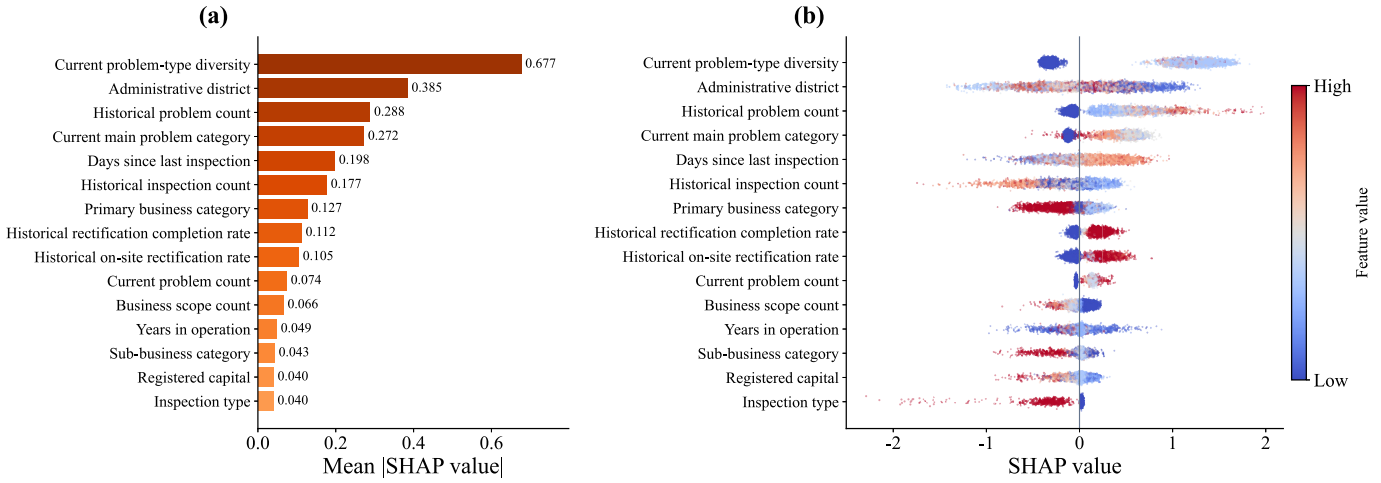


Figure 3. Global SHAP feature importance for inspection-problem prediction. (a) Mean absolute SHAP values for the top 15 features. (b) SHAP beeswarm plot showing the distribution of feature contributions across test-set observations; color indicates feature value (red = high, blue = low). For nominal categorical variables, color reflects encoded numeric values and does not imply ordinal relationships. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

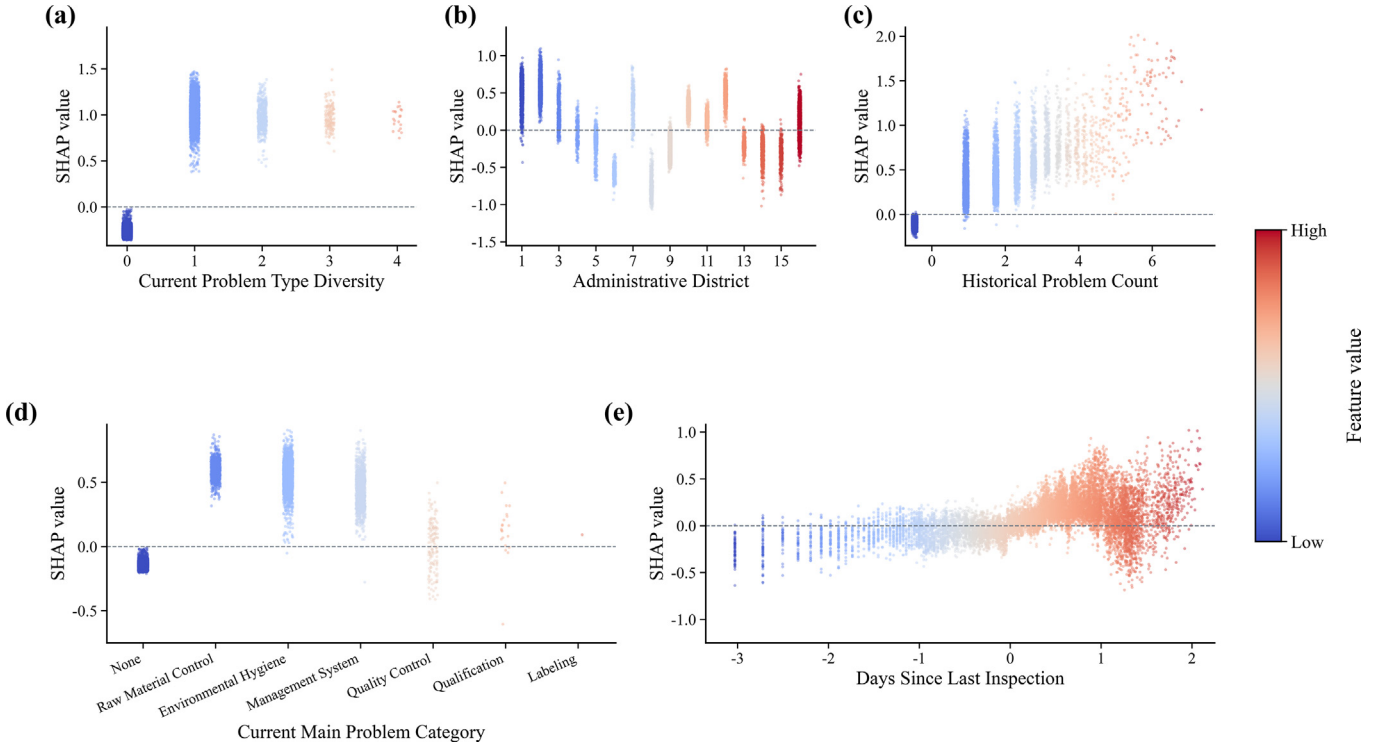


Figure 4. SHAP dependence plots for selected leading predictors. (a) Current problem-type diversity. (b) Administrative district. (c) Historical problem count. (d) Current main problem category. (e) Days since the last inspection. Each point represents one test-set observation; color indicates feature value.

environmental hygiene, indicating that several high-frequency categories tended to converge toward environmental hygiene findings at the next inspection.

The pairwise Lift matrix (Fig. 5b) revealed a complementary pattern. This matrix was calculated at the expanded category-pair level, measuring the association strength between specific antecedent–consequent category pairs. Common operational categories—environmental hygiene, raw material control, and management systems—recurred frequently in absolute terms but showed modest self-transition Lift values (1.37, 1.56, and 1.66, respectively), indicating that their high recurrence was largely explained by their baseline marginal probability. By contrast, rarer compliance categories—qualification compliance, labeling compliance, and quality control—showed much higher self-transition Lift values (14.53, 11.13, and 6.95), indicating disproportionate self-recurrence relative to what would be expected under independence. Significant cross-category Lift values (after Benjamini–Hochberg correction) were observed primarily among compliance-related categories and between compliance categories and environmental hygiene, suggesting that these categories may cooccur within operators with systemic compliance deficiencies.

Persistence rate and persistence Lift provided two complementary observation-level summaries of same-category recurrence (Fig. 5c, d). Unlike the pairwise Lift matrix, which was computed at the expanded category-pair level, persistence Lift was calculated at the observation level, measuring whether an observation containing a given category at the current inspection was disproportionately likely to contain the same category at the subsequent inspection. The two measures differ in unit of analysis and therefore yield different numerical values for the same category's self-recurrence. Persistence rate—the absolute conditional probability of the same category recurring—was highest for environmental hygiene (52.0%), raw material control (38.0%), and management systems (33.6%), reflecting their dominant share of the recurrent problem burden. Persistence Lift—the enrichment of same-category recurrence over the independence baseline at the observation level—was highest for qualification compliance (30.2), labeling compliance (29.3), and quality control (21.8), indicating that once these rarer categories were recorded, they recurred far more often than their baseline marginal probability would predict. Taken together, these results distinguish two patterns: common categories that account for the largest absolute recurrent burden, and rarer categories that show the strongest relative persistence once identified.

GEE sensitivity analyses. GEE sensitivity analyses assessed whether the main continuous and proportion-based SHAP predictors

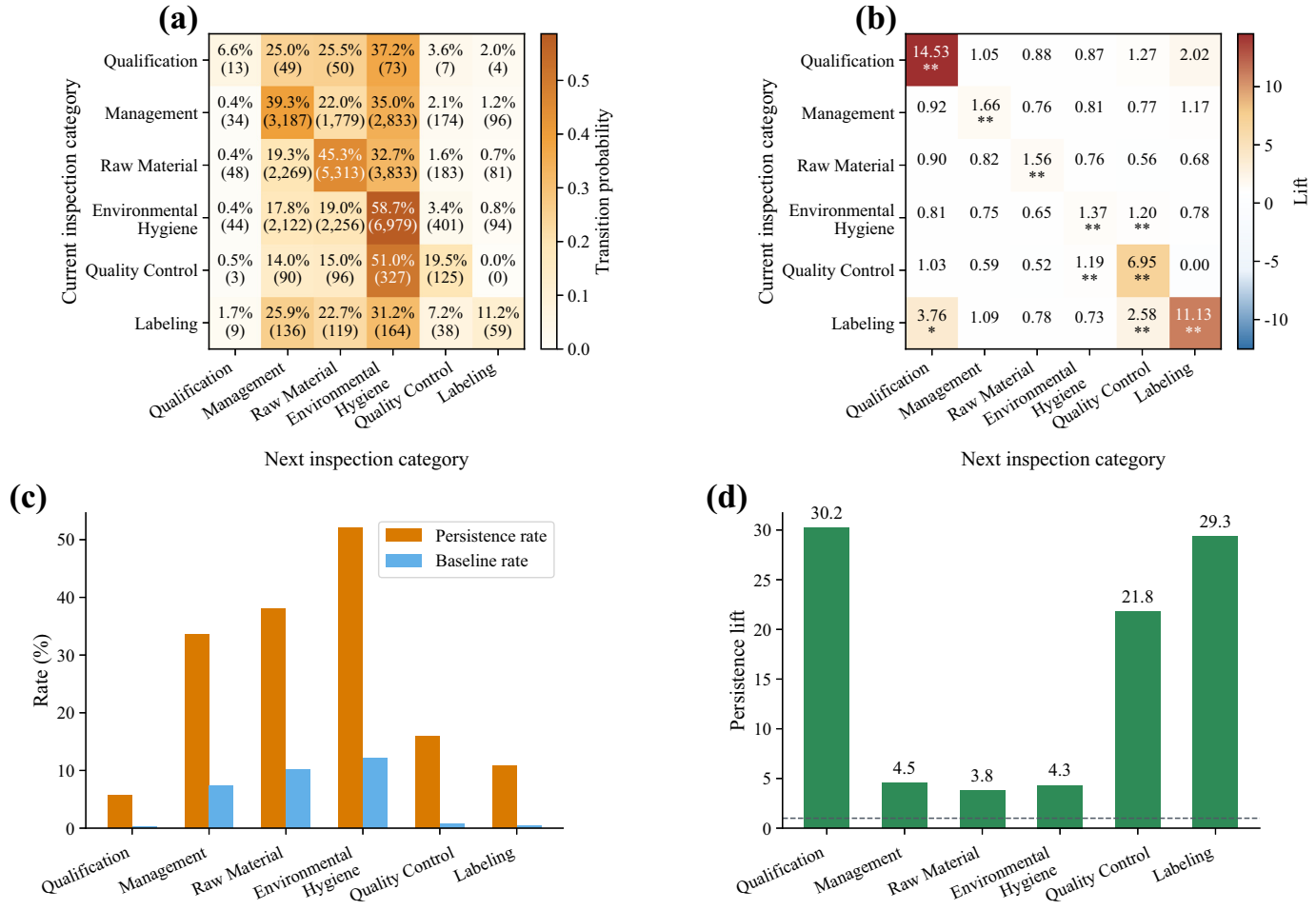


Figure 5. Cross-period persistence and transitions among inspection problem categories. (a) Transition probability matrix showing row-normalized conditional probabilities from each current-inspection category (rows) to each subsequent-inspection category (columns). Cell values show percentages and category-pair counts in parentheses; cell color intensity reflects probability magnitude. (b) Pairwise Lift matrix calculated at the expanded category-pair level; values represent the ratio of observed transition probability to expected probability under independence. Asterisks indicate statistical significance after Benjamini–Hochberg correction ($^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$). (c) Persistence rate (absolute conditional probability of same-category recurrence; orange bars) versus baseline marginal probability (blue bars) for each problem category. (d) Persistence Lift calculated at the observation level; values represent the enrichment of same-category recurrence relative to the independence baseline. The dashed line indicates Lift = 1 (no enrichment beyond independence). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

retained stable directions of association with inspection-detected problems after accounting for repeated observations within enterprises (Table 4).

Of the ten SHAP-identified predictors included in the GEE model, nine showed consistent directional associations across both exchangeable and independence working-correlation specifications. Current problem count, historical problem count, historical rectification completion rate, days since the last inspection, and historical on-site rectification rate were all positively associated with subsequent inspection problems, consistent with SHAP-identified patterns. Among enterprise baseline attributes, registered capital and business scope count showed small negative associations, while years in operation was not statistically significant. One variable showed a directional inconsistency: historical inspection count changed sign across working-correlation specifications (exchangeable OR = 1.015, $P = 0.088$; independence OR = 0.935, $P < 0.001$), indicating sensitivity to the working-correlation specification. Current problem-type diversity showed a negative association in the GEE model despite its strong positive SHAP contribution. This difference may reflect correlation with current problem count and the difference between adjusted associations and contributions to model predictions. The observed inspection interval, included only as a follow-up-duration adjustment, was positively associated with the outcome. Overall, the GEE results showed that the dominant SHAP-identified signals maintained stable directions of association under a parametric framework accounting for within-enterprise correlation.

Discussion

Using 150,467 longitudinal inspection intervals from 46,614 catering service operators in Shanghai, this study examined how routinely collected regulatory data were associated with future inspection-detected problems. Three main findings emerged. First, subsequent inspection problems were strongly associated with time-updated inspection histories, indicating that inspection findings were not isolated administrative events but were partly embedded in the prior regulatory trajectory of each operator. Second, structured complaint and whistleblower records provided complementary but not dominant predictive information. Third, inspection problem categories exhibited distinct temporal recurrence patterns: common operational categories recurred frequently but with modest relative enrichment, whereas rarer compliance-related categories showed much stronger self-persistence relative to their baseline frequency. Together, these findings suggest that risk-based inspection should consider not only the probability of future problems but also the category-specific mechanisms through which problems persist.

In terms of predictive performance, the limited gap between LightGBM and the post hoc weighted ensembles (F1: 0.761 vs. 0.763–0.765) suggests that the main predictive signal was already captured by the strongest single model. LightGBM was therefore retained as the primary model because it combined comparable performance and high recall with an operationally simpler framework; SHAP was subsequently used to explain its predictions.

SHAP analysis indicated that the strongest predictive signals were concentrated in inspection history and timing rather than in complaint or enterprise baseline features. This finding aligns with previous evidence that prior food-service inspection outcomes inform subsequent compliance status (Phillips et al., 2006; Jin & Lee, 2018) and extends it by showing that the predictive value lies not only in whether problems occurred previously but also in their diversity, accumulation, and temporal spacing. In catering service settings, many regulatory problems arise from repeated operational routines—purchasing, storage, cleaning, food preparation, staff supervision, and corrective action. When these routines are weakly institutionalized, problems detected at one inspection may reappear unless the underlying process is changed, producing the strong historical signal observed in the model.

Problem-type diversity and inspection timing each offer distinct mechanistic interpretations. A high number of findings within a single category may reflect a localized failure in one control point, whereas findings distributed across multiple categories suggest that several control points are failing simultaneously. Diversity may therefore function as a proxy for the systemic character of compliance weakness rather than the severity of any single deficiency. The ablation analysis showed that comparable predictive performance was maintained after

Table 4
GEE sensitivity analysis of continuous SHAP-identified predictors with adjustment for inspection interval

Variable	Exchangeable GEE β (SE)	Exchangeable GEE OR (95% CI)	Exchangeable GEE p	Independence GEE β (SE)	Independence GEE OR (95% CI)	Independence GEE p	Direction	Consistent
Current problem-type diversity	−0.409(0.046)	0.664(0.607–0.728)	<0.001***	−0.474(0.054)	0.623(0.560–0.692)	<0.001***	− / −	TRUE
Historical problem count (log)	0.277(0.014)	1.319(1.283–1.356)	<0.001***	0.384(0.016)	1.468(1.422–1.515)	<0.001***	+ / +	TRUE
Days since last inspection (log)	0.152(0.008)	1.164(1.147–1.182)	<0.001***	0.167(0.009)	1.182(1.162–1.202)	<0.001***	+ / +	TRUE
Historical inspection count (log)	0.015(0.009)	1.015(0.998–1.033)	0.088	−0.068(0.009)	0.935(0.918–0.952)	<0.001***	+ / −	FALSE
Historical rectification completion rate	0.187(0.014)	1.206(1.173–1.239)	<0.001***	0.149(0.016)	1.161(1.126–1.197)	<0.001***	+ / +	TRUE
Historical on-site rectification rate	0.098(0.010)	1.103(1.082–1.124)	<0.001***	0.105(0.010)	1.111(1.088–1.134)	<0.001***	+ / +	TRUE
Current problem count (log)	1.179(0.048)	3.253(2.963–3.571)	<0.001***	1.335(0.055)	3.799(3.410–4.234)	<0.001***	+ / +	TRUE
Business scope count	−0.028(0.008)	0.972(0.958–0.987)	<0.001***	−0.030(0.008)	0.970(0.956–0.985)	<0.001***	− / −	TRUE
Years in operation	0.007(0.008)	1.007(0.992–1.023)	0.354	0.008(0.008)	1.008(0.992–1.025)	0.305	+ / +	TRUE
Registered capital (log)	−0.046(0.008)	0.955(0.940–0.971)	<0.001***	−0.051(0.008)	0.950(0.935–0.966)	<0.001***	− / −	TRUE
Inspection interval days (log, adjustment)	0.241(0.008)	1.273(1.252–1.294)	<0.001***	0.265(0.009)	1.303(1.280–1.327)	<0.001***	+ / +	TRUE

Note. GEE models were fitted with the enterprise identifier as the clustering unit, a logit link, binomial variance, and robust sandwich standard errors. Continuous, count, and proportion variables were z-standardized before model fitting; odds ratios correspond to a one-standard-deviation increase. Nominal categorical predictors identified by SHAP (administrative district, current main problem category, primary business category) were excluded because their encoded values do not represent ordinal or continuous changes and cannot be summarized as single odds ratios. Inspection interval days were included as a follow-up-duration adjustment variable only and were not used in the machine-learning models. Direction consistency was assessed by comparing the sign of the coefficient across exchangeable and independence working-correlation specifications. ***$P < 0.001$.

removing this feature, suggesting overlapping predictive information among the remaining predictors. Days since the last inspection captures two overlapping dimensions. On the one hand, longer intervals provide more time for operational conditions to deteriorate between inspections—including staff turnover, procedural relaxation, equipment degradation, and declining management attention—such that problems may accumulate or reemerge before the next regulatory observation. On the other hand, inspection timing also reflects regulatory scheduling decisions: operators perceived as lower-risk may be inspected less frequently, while those with prior problems may receive more frequent visits. These two mechanisms can operate simultaneously and in opposing directions, which is consistent with the nonlinear contribution pattern observed in the SHAP dependence plots. More broadly, these findings are consistent with evidence that food-safety performance in food-service settings is shaped by routine practices, management commitment, and organizational conditions (Powell et al., 2011), and that food-safety risk reflects interacting operational, managerial, and organizational factors (Ze et al., 2024).

The relatively limited predictive contribution of complaint and whistleblower variables can be understood through the different data-generating processes underlying these two information sources. Inspection records arise from standardized regulatory observation according to predefined checklists, whereas complaints originate from consumer exposure, perception, and reporting behavior. Complaint records may therefore capture episodic or experience-based problems that are difficult to observe during scheduled inspections—such as perceived quality failures, foreign objects, or service-process issues—but they are also filtered by consumer awareness, willingness to report, and enterprise visibility. Their predictive value should therefore be interpreted as a consumption-end signal that complements rather than replicates inspection history. This interpretation is consistent with evidence that consumer-generated signals, including online reviews and social media, can supplement public health inspection and outbreak detection (Harrison et al., 2014; Schomberg et al., 2016), and that formal complaint systems, while useful for identifying food-safety events, remain incomplete and behaviorally mediated signals (Mejia et al., 2019; Kim et al., 2024). It also explains the baseline analysis finding that several complaint variables showed slightly higher values in the no-problem group: operators with greater consumer exposure may generate more complaints while simultaneously receiving more regulatory attention, resulting in weaker univariate separation by future inspection outcome.

The cross-period transition analysis revealed that the meaning of recurrence differs across problem categories. Environmental hygiene, raw material control, and management systems had high self-transition probabilities but relatively modest Lift values, indicating that their frequent recurrence was largely proportional to their baseline frequency rather than reflecting strong category-specific persistence. These categories are embedded in daily catering operations and must be maintained continuously through staff behavior, cleaning routines, storage discipline, temperature control, procurement checks, and managerial supervision. Because these tasks are frequent and vulnerable to workload pressure, service demand, and staffing variability, the specific problem item corrected after one inspection may not recur, but a different item within the same operational domain may emerge at the next inspection. This interpretation is consistent with evidence that recurrent food-service violations are common and that sustained food-safety performance depends on routine practices, management commitment, and food-safety culture rather than one-time corrective actions (Phillips et al., 2006; Powell et al., 2011). High recurrence probability with modest Lift therefore reflects a sector-wide burden of maintaining routine operational control rather than strong category-specific persistence.

In contrast, qualification compliance, labeling compliance, and quality control represent a more institutional layer of compliance. These categories depend less on moment-to-moment service behavior and more on formal systems—licensing boundaries, supplier qualification review, label verification, traceability records, quality-control procedures, and internal responsibility assignment. When problems in these areas recur, recurrence is less plausibly explained by routine workload pressure alone; instead, it may indicate that the underlying compliance infrastructure was never adequately established or has structurally deteriorated. The high self-persistence Lift values observed for these categories suggest that once such deficiencies are detected, they recur far more often than their baseline frequency would predict, potentially identifying operators with persistent gaps in governance, documentation, or formal compliance systems. This interpretation aligns with risk-based inspection frameworks that emphasize the role of organizational and procedural conditions as determinants of food-business risk (van Asselt et al., 2021).

These two temporal patterns imply different regulatory responses. For common operational categories with high absolute recurrence, the priority is supporting stable daily implementation rather than simply increasing inspection frequency. Process-oriented interventions—such as standardized rectification checklists, targeted frontline training, and simplified operating procedures—may be more effective than repeated detection of similar problems. For rarer compliance categories with high self-persistence Lift, recurrence signals potential structural deficiencies that are unlikely to resolve through routine follow-up alone. In such cases, closer examination of the operator's formal compliance infrastructure may be warranted, including review of licensing status, supplier documentation, labeling procedures, and internal responsibility systems. This distinction is consistent with risk-based inspection principles that emphasize matching the type and intensity of regulatory intervention to the nature and persistence of the observed risk (World Health Organization Regional Office for the Western Pacific, 2024; Ng et al., 2022).

These findings also connect to the broader literature on food safety culture. The distinction between high-frequency operational recurrence and high-persistence compliance deficiencies may correspond to different layers of food safety culture: the former reflecting the day-to-day implementation of safe practices through routine work, and the latter reflecting the organizational structures that sustain formal compliance over time (Powell et al., 2011; Pai et al., 2024). This perspective helps explain why some problems may resist single-cycle rectification—recurrence may arise from unstable daily implementation or from deeper weaknesses in the systems that maintain compliance responsibilities. However, this interpretation remains inferential, as the present study did not directly measure organizational culture or worker attitudes.

This study has several limitations. First, the outcome was inspection-detected problems rather than a direct measure of food-borne illness or latent enterprise risk. Inspection findings are shaped not only by operator practices but also by inspection type, regulatory priorities, inspector judgment, and special enforcement campaigns; therefore, they represent an administratively mediated proxy rather than a complete measure of food-safety hazard. Second, complaint data are subject to systematic reporting bias. Complaint behavior depends on consumer awareness, platform accessibility, enterprise visibility, and willingness to report, meaning that complaint signals may reflect reporting propensity and consumer exposure patterns rather than underlying hazard alone. Third, the cross-period transition analysis was restricted to inspection pairs where both inspections recorded at least one problem, which represents a subset of operators with repeated problems and may not generalize to the full population of catering service operators. Fourth, SHAP-based interpretation explains model predictions rather than causal effects; observed associations may reflect confounding by unmeasured factors such as management turnover, ownership changes, or local enforcement intensity. Fifth, the analysis was based on data from Shanghai, where regulatory infrastructure, digital complaint channels, and consumer reporting awareness may differ from other regions. Future studies should extend

this approach to multiple cities with diverse regulatory environments and assess whether the identified predictive signals and temporal patterns remain stable across contexts.

Conclusions

This study demonstrates that routinely collected administrative data—including inspection histories, structured complaint and whistleblower records, and problem-category sequences—contain sufficient signal to support forward-looking stratification of catering service operators according to their probability of inspection-detected problems. The predictive model achieved good discrimination using time-updated regulatory features, with inspection history providing the dominant signal and complaint records offering complementary information. Cross-period transition analysis further revealed that recurrent regulatory findings should not be interpreted uniformly: common operational categories reflect the burden of maintaining daily routines, whereas rarer compliance categories exhibit disproportionate self-persistence that may indicate structural deficiencies in formal compliance systems. This distinction may help regulators move beyond general risk ranking toward mechanism-informed follow-up strategies that match intervention type to the nature of the recurring problem.

CRediT authorship contribution statement

Wen-Hui Wang: Writing – original draft, Formal analysis, Conceptualization. **Yi-Qi Li:** Conceptualization. **Yang Guo:** Formal analysis. **Zi-Wen Zhou:** Writing – review & editing. **Shao-Jie Peng:** Writing – review & editing, Supervision.

Funding

This research was funded by the Science and Technology Research Project of the Shanghai Municipal Administration for Market Regulation, grant number 2026KJ03.

Data availability statement

The data that support the findings of this study are not publicly available due to the sensitive nature of the administrative regulatory data. Inquiries regarding data access should be directed to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.jfp.2026.100929.

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